

Seminar: LLM Agents for Survey Questionnaire

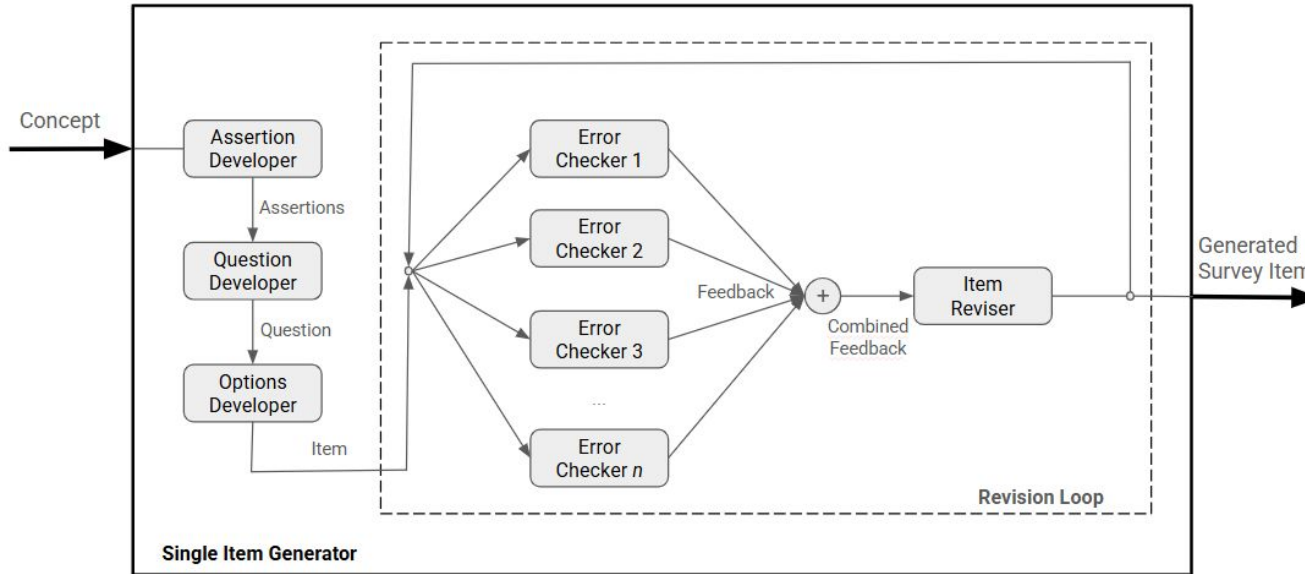
April 13, 2026

Our goal this semester:

We are building an agent pipeline for the development of survey questionnaires!

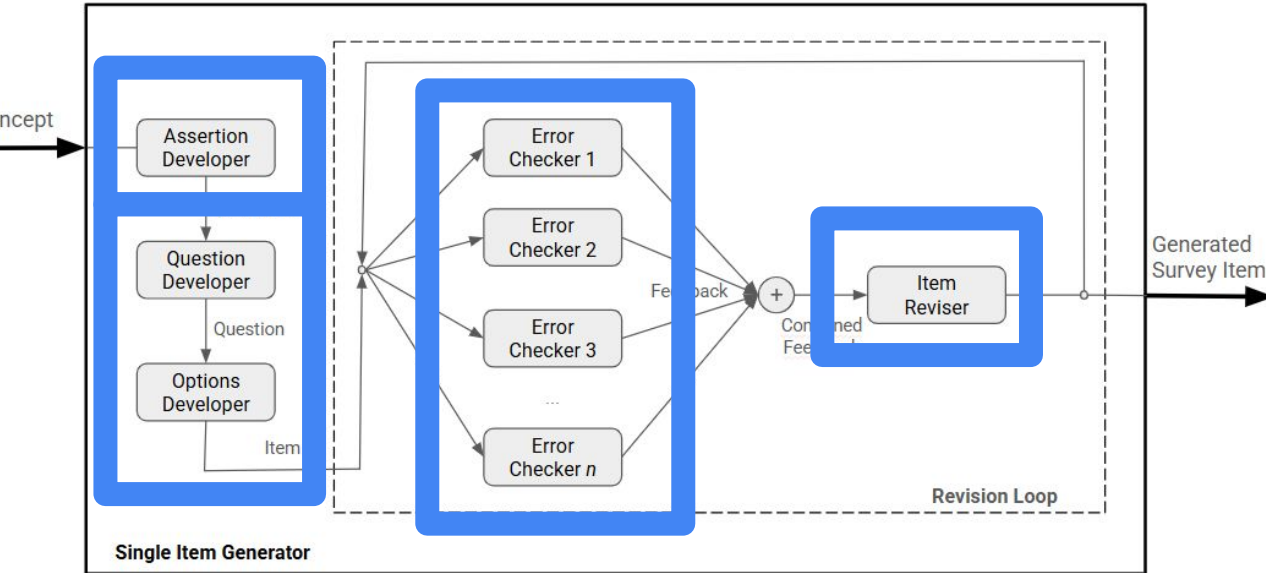
This means that you will learn a lot about

- Surveys
- LLMs in general
- And agents!



Our goal this semester:

We are building an agent pipeline for the development of survey questionnaires!



Every project group will tackle one (or a few smaller steps) of the pipeline.

Main book that we will be using for our seminar



Design, Evaluation, and Analysis of Questionnaires for Survey Research

Author(s): Willem E. Saris, Irmtraud N. Gallhofer

First published: 22 January 2007

Print ISBN: 9780470114957 | Online ISBN: 9780470165195

| DOI: 10.1002/9780470165195

Copyright © 2007 John Wiley & Sons, Inc. All rights reserved.

<https://onlinelibrary.wiley.com/doi/book/10.1002/9780470165195>

Reading for this week: Introduction chapter

 Full Access

Introduction

Book Author(s): [Willem E. Saris](#), [Irmtraud N. Gallhofer](#)

First published: 22 January 2007 | <https://doi.org/10.1002/9780470165195.ch00>

[Go here for SFX](#) 

<https://onlinelibrary.wiley.com/doi/10.1002/9780470165195.ch00>

**Overview of the steps done in developing a survey question.
They don't 100% map to the agents we will be building but
almost!**

Course Schedule

Mondays from 14:00 to 16:00 c.t. weekly in the first half of the semester, afterwards quick check-ins before we meet again for the presentations in the end.

Amalienstr. 52 - K 202

Check the syllabus for an overview of the sessions. We might also update the syllabus during the semester.

Grading

Group project (2 group members, Bolei and Caro will interview you in Week 4 and match you with the agents):

- Presentation in the last two weeks of the semester + Term paper (approx. 15 pages) to be handed in until August 31, 2026.
- Each group will build an agent or a group of agents that are tackling one step of the development steps for survey questionnaire items.
- It is important that you both document the evaluation of your agent carefully and also **delve deep into the literature on survey questionnaire development.**
- When submitting your final term paper, please also include a explanation of each group member's contributions to the work.

Instructors

Caro Haensch, c.haensch@lmu.de

- Background in Missing Data, Survey Statistics and Social Sciences
- Interested in using LLMs **for** Social Sciences



Bolei Ma, bolei.ma@lmu.de

- Background in NLP and Linguistics
- Interested in studying the social factors in language (models), semantics and pragmatics



Guest Instructors

Wiebke Weber (LMU)



Trent Buskirk (Old Dominion University)



Adam Eck (Oberlin College)



Students

Who are you? :)

Your experience with LLMs? And survey (data)?



Gemini



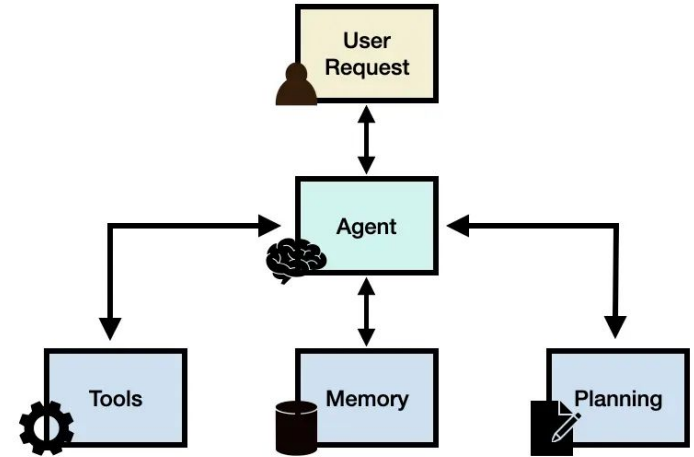
Qwen



deepseek

What is an Agent?

- An agent doesn't just answer questions; it pursues a **goal**.
- It uses an LLM as its core reasoning engine.
- Key Capabilities:
 - User Request: Understanding the environment or the persona it is assigned.
 - Memory: Remembering past interactions or previous answers.
 - Tools: Ability to execute tasks .
 - Planning: Making decisions to achieve an objective without constant human prompting.



What are surveys?

Survey is a systematic method of collecting information from individuals in order to describe, compare, or explain the characteristics, attitudes, or behaviors of a larger population (Groves et al., 2009)

Main Question Types

Choose One

Single-select from a list of options (radio buttons)

Choose Many

Multi-select from a list (checkboxes)

Rating Scale

Rate satisfaction on a 1–10 scale

Ranking

Rank items from most to least preferred

Open-Ended

Free-text: "Why is X your favorite?"

The Topic of This Seminar –

Using LLM Agents for Designing Survey Questionnaires

We explore how multi-agent pipelines (where specialized LLM agents collaborate) can automate and improve the survey design process, from concept mapping to quality checking.

Why Use Agents for Survey Design?

Current Challenges in Survey Design

Writing unbiased, valid questions is time-consuming and requires expert knowledge

Single LLM prompts lack structured quality control and iterative refinement

Errors (double-barreled, leading questions) are hard to catch manually

Translating surveys across languages risks meaning distortion

What Agents Bring

Specialization: Dedicated agents for each task (generation, bias check, translation...)

Iteration: Automated revision loops until quality criteria are met

Scalability: Generate and validate large question banks rapidly

Transparency: Each agent's decisions are traceable and auditable

Possible agents for designing survey questionnaires

Agent 1 — Concept Classification & Indicators Classifies a topic as a concept-by-intuition or concept-by-postulation (Wiebke Weber will walk you the distinction), then derives appropriate indicators. Input: Topic / variable → Output: List of indicators

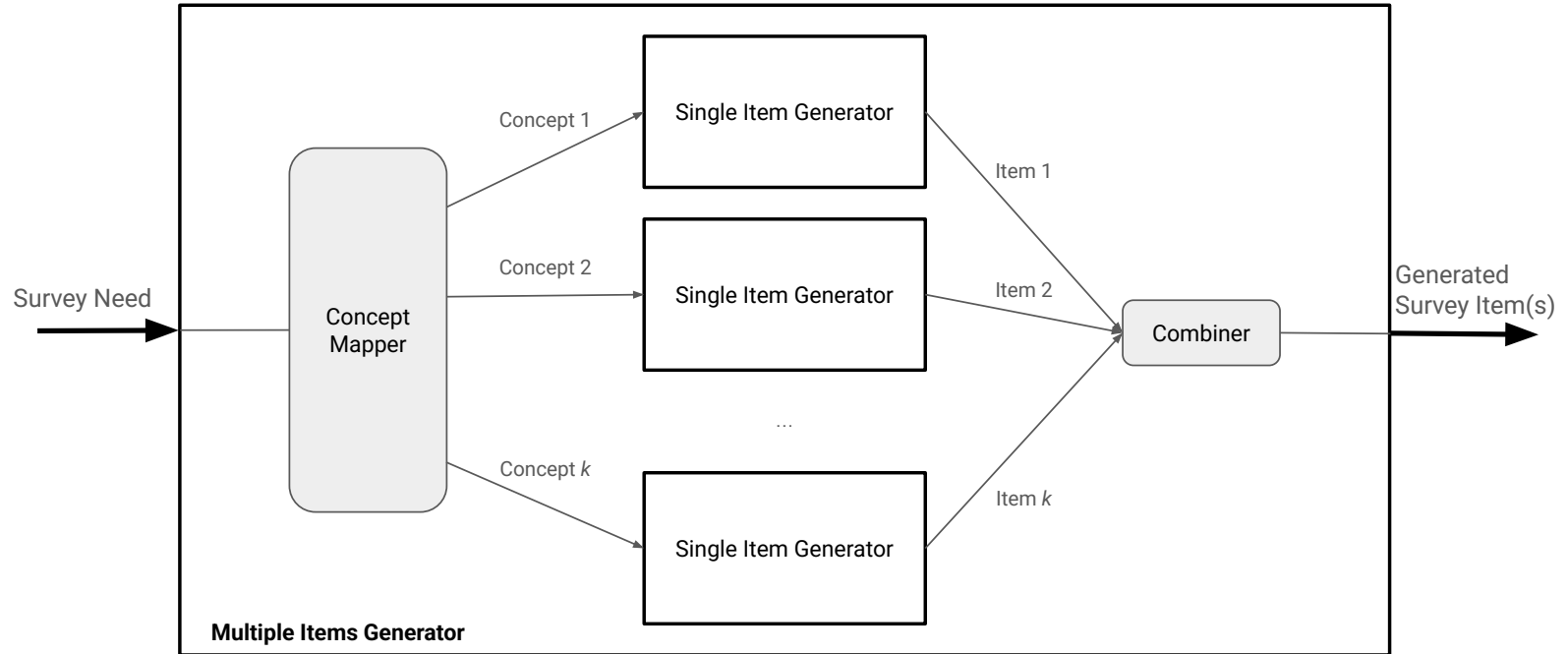
Agent 2 — Assertion Formulation Turns each indicator into a well-formed assertion by selecting the correct basic concept and following the assertion template. Input: Indicators (Agent 1) → Output: One assertion per indicator

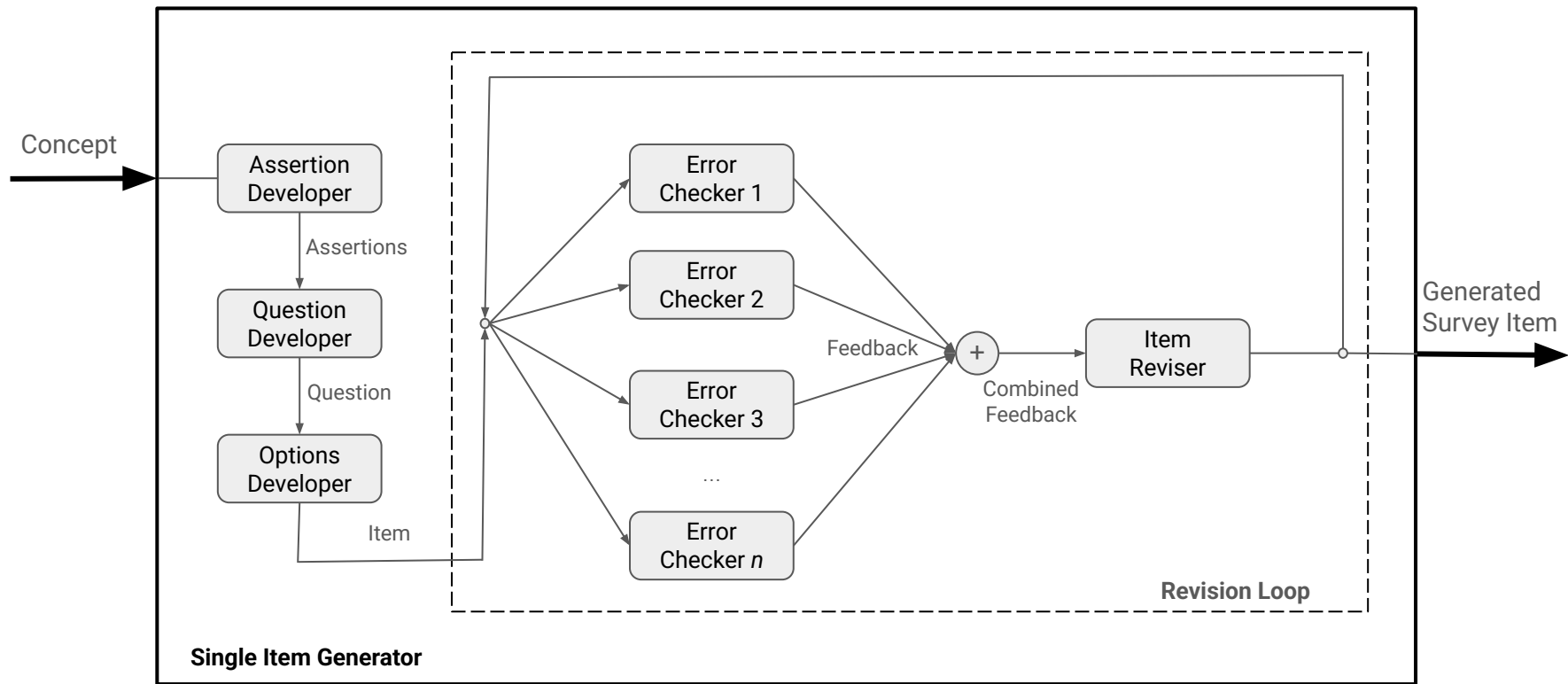
Agent 3 — Question Development Transforms each assertion into a question (direct, indirect, or imperative) including response format and scale. Input: Assertion (Agent 2) → Output: Question incl. answer scale

Agent(s) 4 — Item Quality Check Reviews each question for wording errors (leading questions, vagueness, recall bias) and scale errors (imbalance, polarity, number of points). Input: Question (Agent 3) → Output: Corrected final item

Agent 5 — Item Revision Revises items based on the quality check findings and produces an improved final version. Input: Flagged item (Agent 4) → Output: Revised final item

An Example Agent Pipeline for Survey Design





Preliminary Outline of the Seminar

Date	Topic / Activity	Notes
20 Apr	LLMs with Bolei	
27 Apr	LLMs on the LRZ with Bolei	
04 May	Matching Students to Agent Topics + Interviews (Caro)	<i>Project groups formed</i>
11 May	Questionnaire Design with Wiebke Weber	
18 May	Questionnaire Design with Wiebke Weber	
25 May	PFINGSTMONTAG — No Session	<i>Public holiday</i>
01 Jun	One-on-one meetings on paper topics	
08 Jun	One-on-one meetings on paper topics	
15 Jun	Group check-in on progress	
22 Jun	One-on-one meetings on paper topics	
29 Jun	Interim Presentations	
06 Jul	Interim Presentations	
31 Aug	Submit paper and agent	

Next two sessions

Introductions to LLMs & LRZ system

- Apr 20 — Introduction to Large Language Models (with Bolei)
- Apr 27 — Hands-on with LLMs on the LRZ computing cluster (with Bolei)

Recommended: If you feel your background in NLP could be strengthened, we encourage you to attend the Deep Learning for NLP lecture (Wednesdays 12:00–14:00) by Matthias Aßenmacher in parallel: <https://moodle.lmu.de/enrol/index.php?id=44041>

<https://slds-lmu.github.io/dl4nlp/>

Dont forget the reading for this week: Introduction chapter

 Full Access

Introduction

Book Author(s): [Willem E. Saris](#), [Irmtraud N. Gallhofer](#)

First published: 22 January 2007 | <https://doi.org/10.1002/9780470165195.ch00>

[Go here for SFX](#) 

<https://onlinelibrary.wiley.com/doi/10.1002/9780470165195.ch00>

Overview of the steps done in developing a survey question. They don't 100% map to the agents we will be building but almost!